

Seminar 4: Discrete Random Variables

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1 Probability Distributions and Random Variables

In the previous lectures, probabilities were assigned directly to events in a probability space $(\Omega, \mathcal{F}, P)$. In practice, however, we are often interested in numerical quantities associated with the outcomes of an experiment, such as the number of heads obtained when tossing a coin several times, the number of customers arriving at a store during one hour, or the height of a randomly selected individual.

Such quantities are modeled by **random variables**. To study them, we first introduce probability distributions on the real line.

1.1 Borel σ -Algebra

The **Borel σ -algebra** on $\mathbb{R}$, denoted by $\mathcal{B}(\mathbb{R})$, is the smallest σ -algebra containing all open subsets of $\mathbb{R}$.

Elements of $\mathcal{B}(\mathbb{R})$ are called **Borel sets**. In particular, every open interval, closed interval, half-open interval, countable union, countable intersection, and complement of such sets belongs to $\mathcal{B}(\mathbb{R})$.

1.2 Measurable Functions and Random Variables

Let $(\Omega, \mathcal{F})$ and $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable spaces. A function

$$\xi : \Omega \rightarrow \mathbb{R}$$

is called **measurable** if

$$\xi^{-1}(B) = \{\omega \in \Omega : \xi(\omega) \in B\} \in \mathcal{F} \quad \forall B \in \mathcal{B}(\mathbb{R}).$$

A **random variable** on a probability space $(\Omega, \mathcal{F}, P)$ is simply a measurable function $\xi : \Omega \rightarrow \mathbb{R}$.

We denote by

$$X_\xi = \{\xi(\omega) : \omega \in \Omega\}$$

the set of all possible values of ξ .

If X_ξ is finite or countable, then ξ is called a **discrete random variable**.

1.3 Probability Distributions

A **probability distribution** is any probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

That is,

$$P : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1].$$

If there exists a finite or countable set $X \subseteq \mathbb{R}$ such that $P(X) = 1$, then P is called a **discrete probability distribution**.

1.4 Distribution of a Random Variable

Let ξ be a random variable on $(\Omega, \mathcal{F}, P)$.

The **probability distribution of ξ** is the probability measure P_ξ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ defined by

$$P_\xi(B) = P(\{\omega \in \Omega : \xi(\omega) \in B\}) = P(\xi \in B), \quad B \in \mathcal{B}(\mathbb{R}).$$

Thus, the distribution of a random variable completely describes the probabilities of all events involving its values.

1.5 Probability Mass Function (PMF)

Let ξ be a discrete random variable and let $X_\xi = \{x_1, x_2, \dots\}$.

The **probability mass function (PMF)** of ξ is the function

$$p_\xi : \mathbb{R} \rightarrow [0, 1]$$

defined by

$$p_\xi(x_k) = P(\xi = x_k), \quad x_k \in X_\xi,$$

and $p_\xi(x) = 0$, $x \notin X_\xi$

The PMF completely determines the distribution of a discrete random variable and satisfies $\sum_{x \in X_\xi} p_\xi(x) = 1$.

2 Examples of Discrete Probability Distributions

2.1 Uniform Distribution

Let X be a finite set. The uniform distribution on X is defined by

$$P(x) = \frac{1}{|X|}, \quad x \in X.$$

2.2 Bernoulli Distribution

For $p \in (0, 1)$, $X = \{0, 1\}$, and

$$P(0) = 1 - p, \quad P(1) = p.$$

We write

$$\xi \sim \text{Ber}(p).$$

2.3 Binomial Distribution

For $n \in \mathbb{N}$ and $p \in (0, 1)$, $X = \{0, 1, \dots, n\}$, and

$$P(k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

We write

$$\xi \sim \text{Bin}(n, p).$$

2.4 Geometric Distribution

For $p \in (0, 1)$, $X = \mathbb{N} \setminus \{0\}$, and

$$P(k) = (1 - p)^{k-1} p, \quad k \geq 1.$$

We write

$$\xi \sim \text{Geom}(p).$$

2.5 Poisson Distribution

For $\lambda > 0$, $X = \{0, 1, 2, \dots\}$, and

$$P(k) = \frac{\lambda^k e^{-\lambda}}{k!}.$$

We write

$$\xi \sim \text{Pois}(\lambda).$$

3 Poisson Limit Theorem

Let $\xi_n \sim \text{Bin}(n, p_n)$ and suppose that $np_n \rightarrow \lambda > 0$.

Then for every fixed $k \geq 0$,

$$P(\xi_n = k) \rightarrow \frac{\lambda^k e^{-\lambda}}{k!}.$$

In other words, the binomial distribution converges to the Poisson distribution when the number of trials becomes large and the success probability becomes small in such a way that the expected number of successes remains approximately constant.

4 Independence of Random Variables

Two discrete random variables ξ and η are called **independent** if

$$P(\xi = x, \eta = y) = P(\xi = x)P(\eta = y)$$

for all possible values $x \in X_\xi$ and $y \in X_\eta$.

Similarly, a collection of random variables $\xi_1, \dots, \xi_n$ is called mutually independent if

$$P(\xi_{i_1} = x_{i_1}, \dots, \xi_{i_k} = x_{i_k}) = \prod_{j=1}^k P(\xi_{i_j} = x_{i_j})$$

for every subset $\{i_1, \dots, i_k\} \subseteq \{1, \dots, n\}$.

4.1 Convolution of Discrete Distributions

Let ξ and η be independent discrete random variables. Then

$$P(\xi + \eta = k) = \sum_{i=-\infty}^{+\infty} P(\xi = i)P(\eta = k - i).$$

This formula is called the **convolution formula**.

Important examples:

- If $\xi \sim \text{Bin}(n, p)$ and $\eta \sim \text{Bin}(m, p)$ are independent, then

$$\xi + \eta \sim \text{Bin}(n + m, p).$$

- If $\xi \sim \text{Pois}(\lambda_1)$ and $\eta \sim \text{Pois}(\lambda_2)$ are independent, then

$$\xi + \eta \sim \text{Pois}(\lambda_1 + \lambda_2).$$